Rishikesh Kasireddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	9.42
Sri Chaitanya Junior Kalasala	Class XII (TGBIE)	2023	97.40%
Johnson Grammar High School	Class X (ICSE)	2021	92.00%

SCHOLASTIC ACHIEVEMENTS

Received the **Academic Excellence Award** for outstanding performance in the first academic year **(Issued by IITH)**

Ranked **344 in TS EAMCET** out of 200,000+ candidates.

WORK EXPERIENCE

Microsoft

Software Engineering Intern, Azure Networking

- Built **Zero-Trust device-posture enforcement** for the enterprise Azure VPN Client in C++, blocking non-compliant devices via **10 security checks** (firewall, antivirus, OS version etc)
- Enforced continuous compliance by re-checking posture mid-session and **auto-disconnecting** devices on policy drift
- Added **zero connect-time overhead** by running all checks in a dedicated parallel process during the 4-6 s tunnel setup
- Gave **full visibility** into the feature by logging failed checks and disconnects to Microsoft's Geneva telemetry

Flockshop AI

Machine Learning Intern

- Cut manual editing time from **30-40 min to ~10 s per SKU** with an accessory-replacement pipeline built on Moondream, Flux Fill API and OpenCV masking
- Automated quality checks by training **YOLOv8s and YOLOv8n-pose** models, eliminating hours of manual review
- Built end-to-end data pipelines with Google Sheets and Drive integration for faster, more scalable client delivery

PROJECTS

Multi-Client LLM Quiz System (GitHub)

C++ · TCP Sockets · POSIX Threads · Wireshark

- Built a concurrent TCP client-server quiz system in C++ with **POSIX threads** and LLM-generated questions
- Designed a **custom application-layer protocol** with keep-alive/alive-ok heartbeats for session management
- Reduced LLM calls with a **server-side cache** for popular genres; profiled throughput and latency in Wireshark

RISC-V Pipelined Processor Simulator (GitHub)

C++ · Python · Ripes · Computer Architecture

- Built a cycle-accurate RISC-V simulator with a **5-stage pipeline**, data forwarding and hazard/stall handling
- Implemented static, 1-/2-bit, Perceptron and **Gshare branch predictors** with misprediction analysis
- Validated against **Ripes** via automated tests on matrix multiply, quicksort, binary search and GCD benchmarks

Memory Management System Calls in GemOS (GitHub)

C · OS Kernel · Virtual Memory · Page Tables

- Implemented **mmap, munmap and mprotect** syscalls in the gemOS kernel, tracking VMAs and their protection flags

- Built lazy allocation via a custom **page-fault handler** that sets up page-table entries on first access
- Implemented cfork with **Copy-on-Write** page sharing, deferring physical page copies until a write fault

Per-User Fairness in Flow-Aware Networks (GitHub)

Python · Ryu SDN Controller · Mininet · OpenFlow

- Built a Ryu **SDN controller** implementing per-user fairness (IEEE paper) with admission control and RAMAF
- Achieved a Jain's Fairness Index of 0.99 and **~2.5× link efficiency** over a baseline controller under high load

SKILLS

Languages: C++, C, Python, SQL

Systems: Linux, POSIX Threads, Socket Programming, Multithreading

Tools & Platforms: Windows, Git, Docker, MySQL, Wireshark

ML / AI: TensorFlow, OpenCV, NumPy, Pandas, Matplotlib

RELEVANT COURSES

Core: Programming, Algorithms, Data Structures, Operating Systems, Computer Networks, Database Management Systems, Parallel and Concurrent Programming, Computer Architecture, Machine Learning, Deep Learning

Math: Probability & Random Variables, Statistics, Linear Algebra

LEADERSHIP

Buddy Head, Sunshine Club (Mental Health Club), IIT Hyderabad - Led initiatives promoting student well-being and organised mental health awareness sessions.

Workshop Coordinator, Elan & Vision Fest, IIT Hyderabad - Organized technical and non-technical workshops for one of India's largest student-run festivals